



Predicting and Discovering Weather Patterns in South Africa Using Spatial-Temporal Graph Neural Networks

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Abstract. This study explores Spatial-Temporal Graph Neural Networks (STGNNs) for predicting and discovering weather patterns in South Africa. We compared the predictive performance of three recent STGNNs, i.e. the Adaptive Graph Convolutional Recurrent Network (AGCRN), Conditional Local Convolution Recurrent Network (CLCRN) and Graph WaveNet (GWN) to a Temporal Convolution Network (TCN) to predict temperature, humidity, pressure and wind speed at 45 weather stations in South Africa. We also analysed the quality and usability of the spatial-temporal dependency graph learnt by the different STGNNs. Using the TCN as a baseline we analyse the increase in performance by adding spatial information and identify the stations most likely to be influenced by changes in neighbouring stations. While AGCRN achieved the best overall predictive performance, the dependency graph learned by CLCRN captured the most plausible spatial-temporal dependencies. The findings highlight the strengths and limitations of AGCRN and CLCRN for weather prediction and for automatic discovery of prominent weather patterns in South Africa.

Keywords: Spatial Temporal Graph Neural Networks · Weather prediction · Spatial Temporal Data Mining

1 Introduction

A weather system can be characterised as a highly dynamic and chaotic system that exhibits complex and often latent spatial and temporal dependencies or flows [4]. While traditional deep neural network (DNN) approaches, like the Temporal Convolutional Network (TCN), are able to learn temporal dependencies, they are not designed to deal with spatial dependencies. Spatial-Temporal Graph Neural Networks (STGNNs) [7], an emerging set of DNNs, can deal with both

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temporal and spatial dependencies and generally outperform traditional DNNs on flow prediction applications in dynamic environments.

Formulating the graph structure is an essential aspect of STGNNs. For weather prediction, each weather station is represented as a node in a graph that captures the local weather dynamics in the area around the station, whilst the system level or global dynamics are captured by weighted edges that reflect the strength of the dependencies between stations [4, 10]. Thus, a station (node) will have strong connections (links) to stations that are affected by changes in weather conditions at its location and weaker links to stations that it does not affect. In this way, a node captures local dynamics (single location), while the overall graph structure captures the global (inter-location) dynamics of the system.

While STGNNs have been widely applied for predicting traffic flow at different points in a city traffic network [18], they are less explored for weather modeling and prediction. In a recent study [4] the authors explored two STGNN techniques, GraphWaveNet (GWN), one of the most established STGNNs, with the Low Rank Weighted Graph Neural Network (WGN), and compared these with a Long Short-Term Memory and a TCN for temperature prediction at 21 weather stations in South Africa. Their results showed that GWN outperformed the other techniques including WGN. Newer and more advanced STGNN techniques have appeared since GraphWaveNet. The Conditional Local Convolution Recurrent Network (CLCRN) [10] is a specialised STGNN designed for global scale meteorological prediction and was shown to outperform other STGNNs for coarse grain weather prediction on the global WeatherBench dataset, while the Adaptive Graph Convolutional Recurrent Network (AGCRN) [1] was designed to capture fine-grained node-specific spatial and temporal dependencies and outperforms other STGNNs for city traffic prediction.

In this study we evaluate and compare AGCRN and CLCRN for weather prediction in South Africa. We use a dataset with six weather attributes observed at 45 weather stations and totalling 113 929 observations from the Western Cape, Northern Cape, and Eastern Cape regions of South Africa. We compare the STGNNs for predicting temperature, pressure, humidity and wind speed at these stations over five different time horizons. We performed a systematic node level and diffusion path analysis on the dependency graph and highlight the potential of STGNNs for automatic discovery of weather patterns.

This paper is organised as follows: Sect. 2 describes related work on STGNNs and DNNs for weather prediction; Sect. 3 describes the design of the experiments; Sect. 4 describes the results of the experiments including a comparison of prediction performance, an analysis of the performance at different stations, the dependency analysis and the path analysis; Sect. 5 summarises the key findings; and finally Sect. 6 discusses the limitations of the study and directions for future work.

2 Background and Related Work

2.1 Problem Formulation

We define a graph as:

$$G = (N, E) \quad (1)$$

where N is the set of nodes representing the different weather stations, and E is the set of edges representing the dependencies between stations.

The adjacency matrix $A \in \mathbb{R}^{n \times n}$, where n is the number of weather stations, represents the structure and strength of the dependencies between the nodes (stations) [4]. The observations at a weather station i , is a time series denoted as:

$$X_{it} = \{X_{i1}, X_{i2}, \dots, X_{iT}\} \quad (2)$$

where an observation X_{it} consists of the features observed at the weather station, i.e. wind speed, wind direction, pressure, humidity, rainfall and temperature at each time step, t over a total of T time steps [4].

Given a sequence of historical weather observations for all features, along with the adjacency matrix A , the goal is to predict a series of future values $y_1, y_2, \dots, y_H$ for a given weather feature at each station. The prediction horizon or number of steps to predict is denoted by H [4, 10].

2.2 Spatial-Temporal Graph Neural Networks for Weather Prediction

Spatial-Temporal Graph Neural Networks (STGNNs) have been explored widely for flow prediction in dynamic environments, including traffic prediction [7], stock market prediction [13] and electricity forecasting [15]. Their applicability for weather predictions is starting to be explored. Ma *et al.*, [11] shows how the Hierarchical Spatio-Temporal Graph Neural Network (HiSTGNN) outperforms on real-time meteorological datasets. The study recommends using large-scale data with more variables and stations to enhance forecasting accuracy.

Sáenz et al. [5] apply a Graph Convolutional Network (GCN) combined with a Long Short-Term Memory (LSTM) network for predicting temperatures in Spain, achieving high prediction accuracy and robust performance.

The CLCRN implemented by Lin et al. [10] explores four weather variables and compares their performance against seven baseline models. The results indicate high performance, although there are limitations related to over-smoothness in learned local patterns and instability in the training process.

Similarly, Davidson [4] implements Graph Wave Networks (GWNs) and Weighted Graph Networks (WGNs), which outperform the baseline models, TCN and LSTM. Further, GWN exhibits superior stability and robustness in temperature prediction compared to WGN.

2.3 TCN Architecture

TCNs are deep learning models typically composed of several blocks of dilated causal convolutions, where each convolution layer only considers past sequence values and has no access to future values. The dilation factor regulates the size of each layer's receptive field, allowing TCNs to capture patterns at different time scales [12]. TCN was shown to outperform other temporal DNNs [4] for weather prediction and was therefore selected as our first baseline model.

2.4 GWN Architecture

GraphWaveNet (GWN) is the second baseline model. The architecture combines temporal convolution (Gated TCN) and GCN modules. The GCN module includes a self-adaptive adjacency matrix that can be learned using gradient descent and does not need prior information. The Gated TCN module uses one-dimensional convolutions to capture long-range temporal sequences and avoid the vanishing gradient problem. The GCN module extracts node-level dependencies by aggregating neighbourhood features. GWN has shown to produce strong prediction performance across several domains, including traffic flow prediction, stock market prediction and weather prediction [4, 13, 18].

2.5 AGCRN Architecture

The Adaptive Graph Convolutional Recurrent Network (AGCRN) approach [1] combines the capabilities of Graph Convolutional Networks (GCNs) with the sequential features of Gated Recurrent Units (GRU), and similar to GWN is able to dynamically learn spatial and temporal dependencies. The spatial patterns are captured using the Node Adaptive Parameter Learning (NAPL) module and Data Adaptive Graph Generation (DAGG). The NAPL processes temporal patterns by learning unique patterns and assigning parameters to each node for a specific period. In addition, DAGG computes spatial patterns by learning hidden spatial relationships among nodes and refining diverse and complex spatial patterns.

2.6 CLCRN Architecture

The Conditional Local Convolution Recurrent Network (CLCRN) [10] was designed specifically for meteorological forecasting. The architecture comprises a Conditional Local Convolution (CLC) layer and a Recurrent Neural Network (RNN) layer. The CLC layer captures spatial dependencies by performing convolution operations on specific portions of the input data. The RNN is designed to capture temporal dependencies. It constructs a hidden state using the feature map provided by the CLC layer as input at each step, which is fed back into the RNN layer at the next step. To ensure computational efficiency, a sliding window approach is adopted, enabling convolutions only on specific segments of the input data. CLCRN was shown to outperform state-of-the-art

techniques for weather prediction, specifically the Multi-Scale Spatial Temporal Graph Convolutional Network (MSTGCN), Diffusion Convolutional Recurrent Neural Network (DGCRNN), and the Spatio-Temporal Graph Convolutional Networks (STGCN).

3 Experimental Design

3.1 Data

We use a dataset of weather observations from 70 automatic weather stations across South Africa obtained from the South African Weather Service (SAWS). The dataset contains hourly data for six weather attributes, temperature, wind speed, wind direction, pressure, humidity, and rainfall, over 13 years (2010 to 2022) covering three provinces, the Western Cape, Northern Cape, and Eastern Cape. However, our study focuses on predicting only four of the attributes: temperature, wind speed, humidity, and pressure.

3.2 Pre-processing and Partitioning

We followed the pre-processing approach outlined by Hewage et al. [6] and Davidson et al. [4]. A weather station is only included in the dataset if less than 5% of its data is missing. Of the original 70, 25 stations did not meet this criteria which results in a final dataset of 45 stations, with 15 from each province. The inverse distance weighting (IDW) method was used for imputing missing data at the these stations [16]. Next, the data for the weather features is normalised to the range $[0, 1]$, and then a sliding window is applied to create input/output samples of the time series for each weather feature. Given a time series X , where x_0 represents the current time step, and with window length w and prediction horizon H , the input-output pair of the sliding window technique is represented as $[x_0 - w, \dots, x_0]$ and $[x_1, \dots, x_1 + H]$, respectively.

We make use of the holdout validation technique as our primary validation method, consistent with the approach adopted by several STGNN studies [2, 9, 19]. We divided our 13 years of data into a training set of 11 years (2010 to 2020), 1 year (2021) for validation and 1 year (2022) for testing which gives a 11:1:1 split in terms of years or 84%:8%:8%. Our total dataset contained 113 929 observations, with a training set of 79750 samples, a validation set of 17089 samples, and a test set of 17089 samples for each of the 45 stations.

3.3 Model Implementation and Evaluation

Four different techniques were implemented, a TCN and three STGNN techniques, i.e. GWN, AGCRN and CLCRN. Four independent experiments were conducted to predict temperature, pressure, humidity and wind speed at 3, 6, 9, 12 and 24h horizons. In each experiment, the STGNNs are trained to predict across all 45 stations concurrently, while a dedicated TCN model is created

for each weather station. Hyperparameter optimisation (HPO) is performed for each technique using a random search approach. We generated random combinations of hyperparameters and used each configuration to train the model for the first two years of the training set, but only for a 24-h prediction horizon. The validation loss was calculated for each iteration, and the combination that yielded the lowest loss was chosen as the optimal hyperparameters. A breakdown of the hyperparameters explored and the optimal values found and eventually used in the experiments is shown in Table 1.

Table 1. Parameter Search Space and Optimal Parameters Found for Each Model

Parameter	Search Space	TCN	GWN	CLCRN	AGCRN
# of Epochs	[20, 30, 40]	40	40	40	40
Batch Size	[32, 64, 125, 250]	[32, 64, 125]	64	32	64
Learning Rate	(0.0001, 0.05)	(0.0003, 0.003)	0.01	0.05	0.01
Input Window Size (hours)	[12, 24, 48, 72, 168]	[48, 72, 168]	[24, 48]	12	12
# of Layers	[1, 2, 3, 4]	1	[1, 2]	2	2
Dropout Rate	[0.1, 0.2, 0.3, 0.4]	[0.1, 0.2]	[0.1, ..., 0.5]	-	-
# of hidden units	[30, 40, 50, 60]	[30, 40, 50, 60]	-	16	-
L1, L2 Regularisers	[0.0001, 0.001, 0.01]	[0.001, 0.01]	-	-	-
Layer Normalisation	[True, False]	-	True	True	-
Adjacency Matrix Init.	[Random, Distance]	-	Random	Distance	Random
# of ST layers	[1, 2, 3, 4]	-	1	1	1
# of RNN units	[16, 32, 64]	-	-	32	64
# of Filters	[32, 64, 128]	-	[32, 64, 128]	32	-

We used root mean squared error (RMSE), and mean absolute error (MAE) as commonly employed in the evaluation of STGNNs for prediction tasks [4, 17]. The RMSE, represents the square root of the average squared differences between predicted and actual values, offering a measure of the typical deviation from the observed values. In comparison, MAE calculates the average of the absolute differences between predicted and actual values, providing a more interpretable metric of the average prediction error.

3.4 Experiment Infrastructure

The TCN was implemented in Keras, while GWN, AGCRN and CLCRN was implemented in PyTorch. All experiments were conducted using the facilities provided by the University of Cape Town’s High-Performance Computing Cluster on a A100 GPU or on a Dell XPS 15 laptop with an Intel Core i7-12700H @ 4.70 GHz and an NVIDIA RTX 3050 GPU (4GB GDDR6).

4 Results and Analysis

This section is divided into four main parts. First, we compare the performance of each algorithm for predicting temperature, humidity, pressure, and wind speed.

Next, we analyse the models' performance at the station level, examining how it varied across different stations. Following this, we explore the spatial dependencies highlighted by the adjacency matrices. Lastly, we identify the most prominent dependency paths.

4.1 Predictive Performance

Table 2 shows the average performance of the AGCRN, CLCRN, GWN, and TCN across all 45 weather stations and over different prediction horizons. The MAE and RMSE scores shown are the mean values over three independent runs.

Table 2. Average prediction performance of each algorithm on different weather attributes and horizons. For each variable the best values (lowest) are shown in bold to highlight the best performing algorithm.

Attribute	Metrics	Models	Prediction Horizons					Average	
			3	6	9	12	24		
Temperature	RMSE	TCN	0.03031	0.03793	0.04286	0.04639	0.05486	0.04247	
		GWN	0.03430	0.04408	0.04566	0.04889	0.05295	0.04518	
		AGCRN	0.01982	0.02423	0.02757	0.02988	0.03610	0.02752	
		CLCRN	0.02249	0.02737	0.03024	0.03216	0.03692	0.02983	
	MAE	TCN	0.02102	0.02678	0.03066	0.03333	0.04024	0.03041	
		GWN	0.02435	0.03156	0.03321	0.03647	0.04019	0.03316	
		AGCRN	0.01345	0.01680	0.01927	0.02113	0.02618	0.01937	
		CLCRN	0.01534	0.01912	0.02155	0.02296	0.02673	0.02114	
	Pressure	RMSE	TCN	0.02580	0.03136	0.03567	0.03942	0.05259	0.03697
			GWN	0.01043	0.01201	0.01320	0.01554	0.01832	0.01390
AGCRN			0.00796	0.00918	0.01055	0.01128	0.01420	0.01063	
CLCRN			0.00196	0.00367	0.00393	0.00392	0.00428	0.00355	
MAE		TCN	0.00903	0.01264	0.01592	0.01869	0.02891	0.01704	
		GWN	0.00541	0.00619	0.00678	0.00891	0.01053	0.00756	
		AGCRN	0.00217	0.00274	0.00363	0.00413	0.00644	0.00382	
		CLCRN	0.000732	0.00168	0.00191	0.00187	0.00209	0.00166	
Humidity	RMSE	TCN	0.06687	0.08255	0.09250	0.09961	0.11746	0.09180	
		GWN	0.07682	0.09607	0.10507	0.11192	0.12156	0.10229	
		AGCRN	0.05832	0.07065	0.07902	0.08630	0.10419	0.07970	
		CLCRN	0.06255	0.07586	0.08293	0.08872	0.10467	0.08295	
	MAE	TCN	0.04493	0.05705	0.06486	0.07052	0.08448	0.06437	
		GWN	0.05422	0.06816	0.07514	0.08079	0.08990	0.07364	
		AGCRN	0.03892	0.04807	0.05448	0.05981	0.07285	0.05483	
		CLCRN	0.04152	0.05177	0.05763	0.06189	0.07364	0.05729	
Wind Speed	RMSE	TCN	0.06922	0.07937	0.08529	0.08926	0.09785	0.08420	
		GWN	0.04506	0.05368	0.05732	0.06045	0.06300	0.05590	
		AGCRN	0.03617	0.04105	0.04402	0.04613	0.05217	0.04391	
		CLCRN	0.03826	0.04263	0.04591	0.04850	0.05367	0.04579	
	MAE	TCN	0.05155	0.05952	0.06443	0.06779	0.07506	0.06367	
		GWN	0.03374	0.04051	0.04374	0.04650	0.04957	0.04281	
		AGCRN	0.02633	0.03022	0.03279	0.03446	0.03937	0.03263	
		CLCRN	0.02811	0.03140	0.03405	0.03603	0.04048	0.03401	

From the table, it is evident that pressure exhibits the highest predictive accuracy, followed by temperature and wind speed. In contrast, humidity consistently demonstrates the poorest predictive performance. These findings hold

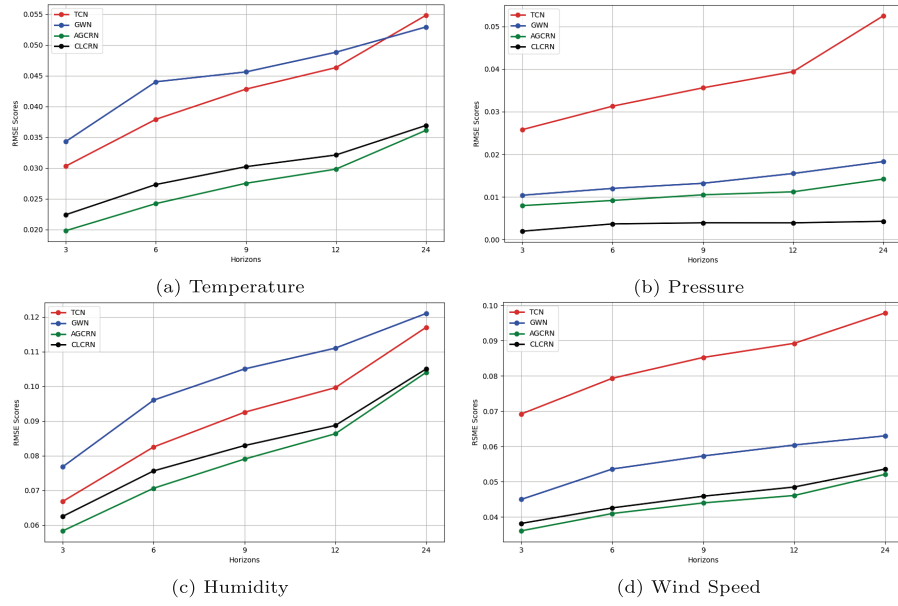


Fig. 1. RMSE values for each model across different forecasting horizons and weather attributes.

consistently across all models, likely due to the inherent differences between these attributes. In terms of predictive performance, AGCRN outperforms the other techniques for temperature, wind speed, and humidity, while CLCRN yields the best performance for pressure. For our baseline models, on average GWN outperformed TCN for the pressure and wind speed variables. However, TCN outperformed GWN on average for the humidity and temperature variables.

We visualise the RMSE values from Table 2 of the different models over different prediction horizons in Fig. 1. As anticipated, all models display decreased accuracy with increasing horizon length. Notably, TCN degrades more rapidly with longer horizons compared to the STGNN models. Although AGCRN outperformed CLCRN for temperature, wind speed, and humidity, Fig. 1 shows that the AGCRN and CLCRN lines lie very close together and are nearly parallel, indicating minimal differences in prediction accuracy across all horizons for these two models. Both models clearly outperform the baseline TCN and GWN models, consistently achieving substantially lower RMSE scores for most weather attributes (Temperature, Humidity and Wind Speed) than the TCN and GWN. The outlier is Pressure, with CLCRN showing better performance across all horizons than AGCRN. AGCRN still outperforms GWN, but not by a large margin.

4.2 Station Level Analysis

For our analysis at the station level we hone in on temperature prediction, the attribute which is predominantly explored in primary studies [4,17] that our work builds upon. We examine and compare the 24h temperature prediction models for the TCN and the STGNNs at each station to better understand the extent to which the incorporation of spatial information impacts on predictive performance.

In Fig. 2(a), the RMSE values of the TCN, CLCRN and AGCRN for 24h temperature prediction are overlaid on a geographic map. Figure 2(b) plots the difference between the RMSE scores for each STGNN to the TCN at each station ($RMSE_{TCN} - RMSE_{STGNN}$). A value of zero indicates the same RMSE value as the TCN, while values above zero indicate an improvement in prediction performance over the TCN. We also plotted the average improvement achieved across all stations to better identify stations where there was a substantial improvement in performance. From this, we can see which stations had the most significant improvements over the purely temporal TCN and where they are situated on the map. We observe that our STGNN models generally outperform the purely temporal TCN model, with the most significant improvements at stations 13 and 22, while the improvement over the TCN is the least pronounced at station 37. In general, we notice that the biggest improvements over the TCN occur mostly at stations along the coast, whereas the least improvements are observed at more inland stations. This suggests that the additional spatial information utilized by the STGNN architecture provides more significant benefits for coastal stations.

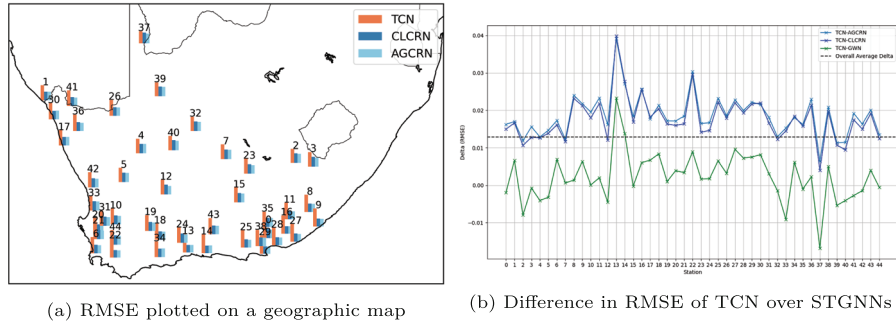


Fig. 2. Comparisons of RMSE scores achieved by each model for temperature predictions on the 24-h horizon at each station.

While the GWN improvement over TCN is less pronounced compared to CLCRN and AGCRN, with some stations where the TCN performed better, it is interesting to note that the relative performance between stations follows a similar pattern to the other STGNNs. For stations where CLCRN and AGCRN show

the greatest improvement over TCN, GWN also demonstrates the most substantial relative improvement. Conversely, for stations where CLCRN and AGCRN show less improvement, GWN shows a similar relative performance. This consistent trend in relative performance to the TCN across different stations is interesting and indicates which stations are most impacted by changes at neighbouring stations.

Given that AGCRN and CLCRN performed similarly, achieving nearly identical RMSE values for temperature, we decided to investigate these two models' predictive results at the station level. Figure 3 plots the actual values in the test set against the values predicted by each model for station 13.

Although this visual is specific to station 13, similar patterns were observed for other stations. From the observations, CLCRN tends to predict higher or more extreme peaks more accurately, while AGCRN generally provides more accurate predictions for average peaks. This indicates that CLCRN is better at detecting extreme temperatures, whereas AGCRN performs better on average temperature predictions. The two bottom plots in Fig. 3 show zoomed-in sections: Section A from March 10 to March 24, 2021, and Section B from May 19 to June 7, 2021. Section A includes average and extreme values, where CLCRN more accurately predicts the highest peaks, while AGCRN is closer to the actual values for lower average peaks. Section B contains fewer extreme values, but the same pattern is observed, with CLCRN excelling in predicting extreme values and AGCRN performing better for average values.

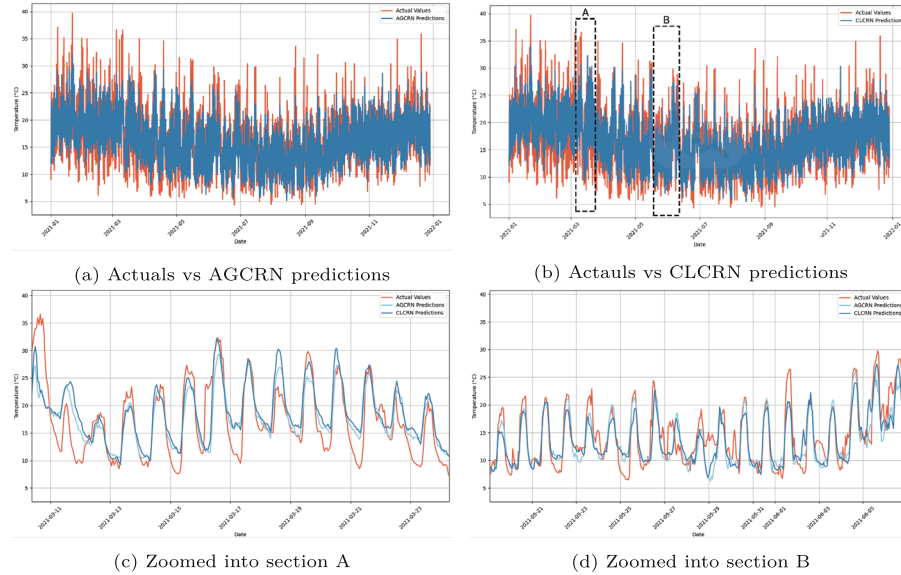


Fig. 3. The actuals and predicted values for AGCRN and CLCRN models at station 13.

4.3 Dependency Analysis

In this section, we analyse the adjacency matrices generated by each model for predicting temperature over a 24-h horizon. It is important to note that AGCRN’s matrix does not rely on any pre-existing knowledge for initialization, as it starts with random initialization. Consequently, any observed patterns in the final matrix of this model are solely the result of the model’s self-learning processes. In contrast, CLCRN’s matrix is initialised by leveraging the K-nearest neighbours algorithm, based on the induced spherical distance calculated from the spatial locations of the weather stations.

First, Fig. 4 represents heat maps depicting the matrices produced by each model, which illustrates the significant differences in how each model utilises and learns from the matrices.

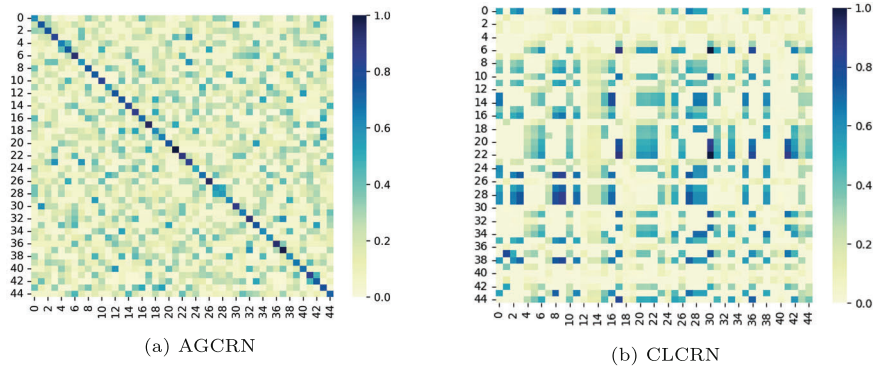


Fig. 4. Heatmaps of the adjacency matrices for 24-h temperature predictions.

The heatmap for the AGCRN model shows a clear emphasis on the diagonal, indicating that AGCRN prioritises each station’s own past information when predicting their future values. Despite this self-emphasis, there are still notable values off the diagonal, distributed relatively evenly across the matrix. This suggests that AGCRN does not assign overriding importance to any single station, as no specific row or column in the heatmap stands out significantly darker than the others. In contrast, the heatmap for CLCRN displays a distinct pattern with a higher prevalence of influential stations and/or stations that are more dependent on others, evident from columns that are noticeably more dominant. Additionally, there is a marked reduction in self-referential dependencies for individual stations compared to AGCRN. This indicates that CLCRN’s modelling approach places greater emphasis on inter-station relationships.

Next, in Fig. 5, we plotted the strongest edges from each matrix onto a map to understand the spatial relationships between the most prominent dependencies. An edge from one station to another indicates that the station receiving the arrow is dependent on the station from which the arrow originates. We excluded

the diagonal (self-referencing) edges and calculated the 98th percentile of the strongest edges to identify the top 40 edges for each model. The 98th percentile for AGCRN was 0.7592, while for CLCRN, it was 0.7039. The comparative analysis of CLCRN and AGCRN reveals significant differences in their dependency structures.

For AGCRN, the strongest dependencies predominantly point to nearby stations, highlighting that a station primarily influences its immediate neighbours. This is particularly noteworthy since station location data was not provided to the model, suggesting that AGCRN learned about its nearest nodes independently. This observation aligns with AGCRN’s architecture, where the NAPL module provides unique parameter spaces for each node, allowing it to capture local patterns more effectively rather than network-wide effects. The presence of bidirectional edges is also evident, indicating that two stations exert mutual influence on each other.

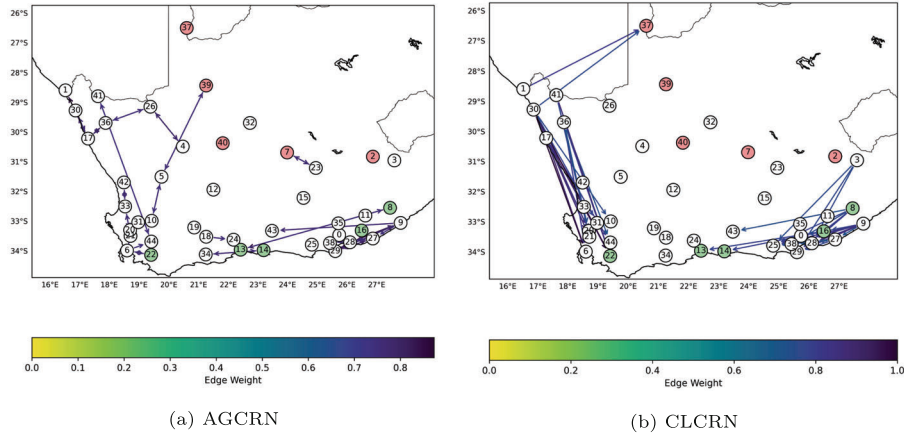


Fig. 5. Strongest dependencies maps drawn from the adjacency matrices for AGCRN and CLCRN models for 24-h temperature predictions.

Additionally, strong dependencies are minimal inland, with most concentrated along the coast. In contrast, CLCRN shows a more distinct pattern in its strongest dependencies. Similar to AGCRN, CLCRN exhibits fewer strong dependencies inland. However, CLCRN’s strong dependencies along the coasts are more pronounced and follow a consistent pattern, extending southward along both the eastern and western coasts.

We then investigate whether stations with strong dependencies exhibit a direct correlation with improved prediction accuracy. To do this, we compare our model to the purely temporal TCN, which has no dependencies. From Fig. 2(b), we identify the top 5 stations with the greatest improvements over TCN (stations 13, 22, 14, 16, and 8) and the 5 stations with the least improvements (stations 37, 39, 40, 2, and 7). Accordingly, in Fig. 5, we have shaded the top 5

stations green and the bottom 5 stations red. As previously noted, STGNN models perform particularly well compared to the TCN model along coastal areas. Interestingly, the stations with the greatest improvements (green) are located along the coast, where dependencies are more prominent. In contrast, more inland areas, where strong dependencies are less prevalent, show less improvement over TCN, with more red stations. For both CLCRN and AGCRN, four out of the five top-improved stations have incoming edges. In contrast, four out of five of the least improved stations for CLCRN have no incoming edges. This suggests that stations receiving strong influential dependencies demonstrate the best improvement and utilization of the STGNN architecture compared to the purely temporal TCN. However, it is worth noting that station 37, which was the station with the least improvement over TCN, did not align with this general pattern as it has two strong incoming dependencies. Therefore, we should be aware that not all dependencies can be said to be true dependencies, and some occurrences may deviate from the pattern.

4.4 Path Analysis

Figure 6 shows the paths found by CLCRN. We lowered the threshold for edge weight to 0.25, and only show paths or chains of at least 3 nodes, i.e. a source node must influence at least one node via another node to be included. This analysis yields some interesting insights into the dynamics of the system.

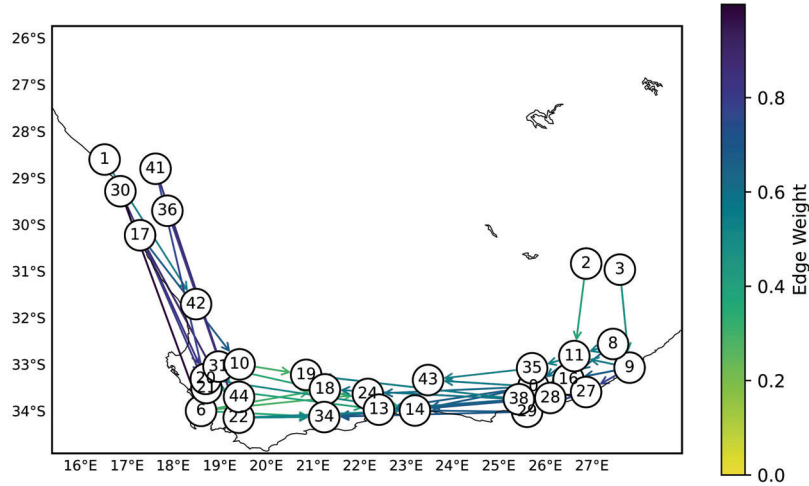


Fig. 6. Visualising Spatial-Temporal Dependency Paths in CLCRN Model for 24-h Temperature Prediction.

Easternmost Stations’ Influence. In terms of the most prominent paths, Fig. 6 shows that there is a strong path of influence originating from a cluster of four stations 2, 3, 8 and 9 in the east.

These stations influence stations to the south and west along a path to a cluster of five stations (35, 0, 38, 28 and 29) on the south east coast. These stations in turn influence their western neighbours stations 13, 14, and 24, which in turn influence stations 18, 19, and 34. Stations 18, 19 and 34 appear to fall at the centre of the dependency network. This sequence illustrates a clear path of spatial-temporal dependencies, from the eastern stations towards and along the southern coast to the centre of the Western Cape.

Northernmost Stations’ Influence. A similar prominent path along the western coast emerges originating from this cluster of northern coastal stations (1, 30, 41, 17 and 36). They exert a strong influence in a southern path along the coast through station 42 to a cluster of seven stations in the south west. The path of dependencies traced from stations 1, 30, 41, 36, and 17, extending down to 31, 20, 21, 22, 6, 10, 44, and finally to 19, 18, and 34, reveals a significant influence pathway. This dependency chain indicates that the climatic conditions at the areas where these five northernmost coastal stations are located plays a crucial role in influencing climatic conditions along the coast.

Convergence of Paths. A noteworthy observation is the convergence of these two critical paths at stations 18, 19, and 34, where the influence from the eastern and north west stations converge. This convergence underscores the central role these stations play in the network, acting as key nodes where the influence from both the easternmost and northernmost stations interconnects. A more detailed analysis of the results suggests that this pattern may be linked to the geographical terrain of the study area. Further research corroborates this hypothesis, pointing to the Great Escarpment as a potential contributing factor [3]. This aligns with a previous study [8] that highlighted distinct weather patterns in the coastal regions and Cape Fold mountains compared to the central regions of South Africa. Further substantiating this hypothesis, our analysis of Fig. 4 and comprehensive examination of the per station dependencies reveal a distinct characteristic among Stations 4, 12, and 40, which are positioned centrally within the spatial map of our data. These stations display minimal influence on other stations and limited dependence on their neighbouring counterparts. Such observations could be attributed to the relative scarcity of nearby weather stations or specific geographical factors unique to these regions.

5 Conclusions

In this study we evaluated two recently developed Spatial-Temporal Graph Neural Network (STGNN) techniques, i.e. the Adaptive Graph Convolutional Recurrent Network (AGCRN) and the Convolutional Long Short-Term Memory Graph

Neural Network (CLCRN) for weather prediction and knowledge discovery in South Africa. This work builds on and expands on the study done by Davidson and Moodley [4]. Their study showed that GraphWaveNet achieved the best prediction performance for temperature prediction on 21 weather stations in South Africa.

Our study used a much larger dataset (45 stations) and also explored STGNNs for predicting three other weather attributes, i.e. pressure, humidity and wind speed, in addition to temperature. We found that the newer STGNNs, AGCRN and CLCRN, outperformed GraphWaveNet, with AGCRN achieving the best overall performance. Moreover, we also explored the efficacy of STGNNs for scientific knowledge discovery. We used the TCN performance as a baseline to identify the effect of adding spatial information, which allowed us to further analyse the dependency graph learnt by the STGNNs. Through visual analysis of the dependency graph learnt by CLCRN and systematic path analysis we were able to discover complex spatial temporal dependency chains which drive the dynamics of the weather system in South Africa.

It must be noted that the approach does not require extensive prior domain knowledge. The adjacency matrix for AGCRN was initialised randomly, while the adjacency matrix for CLCRN was initialised using the induced spherical distance calculated from the spatial locations and leveraging the K-nearest neighbours algorithm (stations). The STGNNs adjust the adjacency matrix during training, so the dependency graphs are automatically learnt from the observational data and the techniques do not require prior domain knowledge.

We believe that this result will have significant impact on researchers in climate science. Not only do STGNNs achieve superior predictive performance, but they can also be used to discover complex weather patterns based purely on observational data. While the study focused on weather data, the approach has broader applicability and can potentially be applied for automatic discovery and explanation of spatial-temporal patterns in other complex and dynamic systems.

6 Limitations and Future Work

This research uncovers multiple avenues for further exploration. Firstly to explore how the prominent dependencies and paths shown in Figs. 5 and 6 align with known theories about the South African weather system. Since mountains are known to have a substantial effect on weather patterns, one immediate avenue for further visual analysis is to overlay these results on a topological map to understand the influence of mountains on the paths.

Another avenue for future exploration could focus on implementing the approach on a global-scale dataset, like WeatherBench [14], to extract dependencies and paths. This would allow for a more comprehensive evaluation of the approach to discover global weather patterns.

Lastly, given that weather predictions often require longer prediction horizons, future work could investigate the applicability of the AGCRN and CLCRN for long term weather prediction beyond 24 h.

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